Why This World, in This Form?

Representation, Cognition, and the Evolution of Human Knowledge

When I was in high school, I often questioned why we had to learn certain things. Why did I have to memorize mathematical formulas or study subjects that did not interest me? What was the purpose of learning knowledge that other people had already discovered? At first, these seemed like ordinary frustrations with education. However, the more I thought about them, the more my questions expanded. Why is our knowledge organized in this particular way? Why has human civilization developed into the world we know today, with these scientific theories, mathematical systems, and social structures? If the earliest civilizations had developed differently, would we still have arrived at the same understanding of the world?

Sometimes, I even imagined parallel universes in which human civilizations followed entirely different paths. Perhaps people in another universe would have developed different mathematical systems, different ways of organizing society, or different approaches to understanding nature. Would their knowledge eventually converge with ours, or might they discover possibilities we have never imagined?

At the time, I did not have a framework for thinking about these questions. After learning about representation, computation, and constraints in cognitive science, however, I began to see them differently. I realized that I was not simply questioning what humans know, but how our ways of representing knowledge influence what we are able to know next.

I have come to think that human knowledge develops through an iterative relationship between representation and cognition. Existing representations allow us to perform new cognitive operations, which may produce new discoveries and representations. These then become the starting points for future generations. Through this process, knowledge accumulates, but it may also become constrained by the structures it has inherited. **I want to explore whether human civilization represents an inevitable convergence toward certain truths, or one of many possible outcomes shaped by the cognitive tools and historical paths through which those truths were discovered.**

Does Truth Have to Look the Way We Represent It?

Mathematics provides an interesting starting point for this question.

Consider the Pythagorean theorem:

$a^2 + b^2 = c^2$

We learn this formula as a mathematical truth about right triangles in Euclidean geometry. Its validity does not depend on whether I personally understand it, and its underlying relationship is not changed simply by writing it differently.

Yet why do we express this relationship using these particular symbols? Why do we use letters, exponents, and equations arranged in this form? We could communicate the same relationship through a geometric construction, a verbal description, or another mathematical notation.

Even if we assume that mathematical truths exist independently of human cognition, it does not follow that the representations through which we understand them are inevitable.

This distinction fascinates me because it suggests that reality and our representations of reality are not the same thing. Physical phenomena existed before humans developed scientific theories to describe them. However, scientific theories are not simply reality copied into language. They involve concepts, measurements, models, and assumptions through which humans interpret observations.

This raises a question: **If different representations can describe the same underlying reality, might they also lead us toward different discoveries?**

Cognitive science provides a way to approach this problem. In Marr's (1982) framework, the computational level concerns what problem a system solves and why, while the algorithmic and representational level concerns how information is represented and transformed to solve that problem. Different representations and algorithms can sometimes accomplish the same computational goal.

For example, consider multiplying 12 by 13. Representing these quantities as Arabic numerals makes it relatively straightforward to apply the familiar multiplication algorithm. Representing the same quantities as Roman numerals, XII and XIII, does not naturally support that particular procedure. The mathematical relationship remains unchanged, but the representation affects the cognitive operations available to us.

This example connects with Zhang and Norman's (1994) work on distributed representations, which examines how external representations participate in cognitive tasks. Representations are not merely passive containers for information. Their structures can make some operations easier to perform than others.

This changed how I understood my original question. Perhaps mathematical formulas are not simply things we memorize after someone discovers a truth. They are also cognitive tools that make certain kinds of reasoning possible.

And if representations shape how we reason, they may also shape what we are likely to discover.

Knowledge as an Iterative Cognitive Process

This leads me to another question: how does human knowledge develop across generations?

We rarely begin learning from nothing. Before I can study calculus, I need some understanding of numbers, algebra, and functions. Before learning a programming language, I must understand

certain concepts about variables, operations, and logical structures. Much of what I learn depends on representations developed by people who lived long before me.

I began imagining human knowledge as an iterative feedback process:

Existing representations → Cognitive processing → New representations → Further cognitive processing

In this process, representations enable computation, and computation can generate new representations. Here, I use computation in the broad cognitive sense of rule-governed information processing, rather than referring only to the operations of digital computers.

A mathematical notation allows someone to perform calculations that might otherwise be difficult. Those calculations can support a new theorem or model. The model then becomes another representation that future thinkers can learn and use.

If we call the existing system of representations R, then each new stage can be thought of as depending partly on the representations available at the previous stage. However, the process also depends on new experiences, evidence, and interactions with the environment. It is therefore not a closed loop in which knowledge simply reproduces itself.

This idea connects with cumulative cultural evolution. Tomasello (1999) describes how human cultural learning allows knowledge and practices to accumulate across generations. Rather than independently rediscovering every tool or technique, individuals can inherit what others have developed and build upon it.

What interests me is that we may inherit more than knowledge itself. **We also inherit the cognitive tools through which knowledge becomes understandable.**

For example, mathematical notation does not simply preserve discoveries from the past. It helps determine how later students learn to manipulate mathematical relationships. Similarly, scientific classifications do not merely organize what is already known. They may influence which similarities researchers notice and which questions they consider worth asking.

In this sense, the representations produced by one generation become part of the cognitive environment of the next.

However, this raises a paradox. If existing representations make new thinking possible, could they also limit the directions in which thinking develops?

When Knowledge Becomes a Constraint

One of the ideas I found most interesting in cognitive science is that complex systems do not have unlimited possibilities. Their behavior is shaped by constraints.

In Marr's framework, the physical implementation of a cognitive system constrains which algorithms can realistically be carried out. At the same time, the computational problem helps explain why certain forms of information processing are useful. These different levels of explanation cannot simply be reduced to one another.

I think a related, although distinct, kind of constraint emerges through the historical development of knowledge.

When a representation becomes widely accepted, it may influence education, research methods, technologies, and subsequent theories. Later generations are trained to work within that framework. Over time, certain ways of describing the world may become so familiar that alternative ways are difficult even to imagine.

This is not exactly what Marr means by constraints from below, since historical and cultural influences operate differently from physical implementation constraints. Nevertheless, both raise a similar question: how do existing structures restrict or enable the possible operations of a system?

I find this particularly interesting when thinking about scientific progress. We often imagine that discovering more knowledge automatically makes our understanding of the world more complete. Yet an established framework might also encourage us to interpret unfamiliar phenomena through concepts we already possess.

For instance, if a scientist encounters an observation that does not fit an existing model, the difficulty may not simply be a lack of information. The available conceptual framework might also be inadequate for describing the phenomenon. Developing a new representation could reveal relationships that were previously difficult to recognize.

Of course, this does not mean that older representations necessarily prevent innovation. Scientific progress often involves revising, extending, or replacing existing models. In fact, understanding an old framework may provide exactly the tools needed to challenge it.

This tension seems central to learning itself: **knowledge expands what we can think, while also influencing which possibilities seem natural or reasonable to consider.**

Perhaps the purpose of education is therefore not only to acquire existing representations, but eventually to recognize their limitations.

Could Civilization Have Developed Differently?

This brings me back to the question I began asking in high school.

If human civilization could begin again, would we eventually develop the same knowledge systems?

Imagine two civilizations that emerge independently, without exchanging information. They inhabit environments governed by the same physical laws and possess similar basic cognitive capacities. However, their early languages, mathematical symbols, cultural practices, and methods of organizing knowledge develop differently.

Would these civilizations eventually discover the same mathematical relationships? Would they develop equivalent scientific theories? Or would their different starting points lead them toward entirely different ways of understanding the world?

I think this thought experiment reveals an important distinction between **convergence and historical contingency**.

On one hand, the shared structure of reality may constrain the kinds of discoveries different civilizations can make. If both civilizations investigate the same physical phenomena, some successful explanations may need to capture similar underlying relationships. In that sense, knowledge might converge toward certain truths.

On the other hand, the representations used to express those truths need not be identical. Different civilizations could develop different notations, computational methods, conceptual categories, or educational traditions.

Even if their knowledge became equivalent in some respects, the historical processes leading to it might remain different.

This suggests that we should distinguish between convergence in what is discovered and convergence in how discoveries are represented.

I do not know whether these civilizations would ultimately converge or diverge. Perhaps both processes would occur simultaneously. Shared environmental constraints could encourage similar solutions, while cultural differences and earlier representations could sustain different intellectual traditions.

My point is not that today's civilization is arbitrary, or that any imaginable knowledge system would work equally well. Reality places genuine constraints on which theories and technologies succeed. Rather, I question whether those constraints are sufficient to explain the particular form our knowledge has taken.

The world might impose limits on what can be true without completely determining how humans come to understand it.

What Happens When AI Enters the Feedback Loop?

Recently, this question has become more personal because of how frequently I use artificial intelligence while learning.

I often have ideas that I find difficult to express clearly or connect to concepts I have already learned. When I discuss them with AI, it can help me organize my thoughts, introduce relevant theories, or suggest connections I had not considered. I then evaluate those suggestions, disagree with some of them, revise my interpretation, and continue the conversation.

This feels like another form of iterative feedback.

My initial representations influence the questions I ask. The AI generates new explanations or possible connections. These outputs influence my subsequent thinking, which produces new questions and interpretations.

However, AI introduces an interesting complication to the development of knowledge. Earlier generations primarily inherited representations through education, books, and social interaction. Now, AI systems trained on large collections of existing human-produced information can participate directly in the process through which people organize and develop ideas.

I wonder what this means for the future of human knowledge.

If increasingly large numbers of people use similar AI systems to study, write, and solve problems, could our ways of representing knowledge become more similar over time?

For example, if AI repeatedly recommends certain established frameworks for interpreting unfamiliar ideas, users may increasingly adopt those frameworks. This could encourage convergence in the questions people ask and the explanations they produce.

Yet the opposite is also possible. AI might expose individuals to unfamiliar perspectives, help them connect previously separated disciplines, and make alternative representations easier to explore. Instead of reducing diversity, it could expand the range of ideas accessible to each person.

These possibilities are not mutually exclusive. AI could encourage convergence in some forms of expression while increasing exploration in others.

I find this especially interesting because I can observe both possibilities in my own learning. Sometimes AI organizes my ideas into familiar academic categories, making them easier to communicate but potentially narrowing my original interpretation. At other times, it introduces connections that allow me to develop the idea further than I could have independently.

I am therefore becoming more aware that the tools I use to think can influence the direction of my thinking.

The same question that applies to mathematical notation and inherited scientific theories may now apply to AI: **does a cognitive tool merely help us express what we already think, or does it also change what we become capable of thinking?**

Conclusion: Learning to Question the Representations We Inherit

Looking back, I think my high school questions were less about disliking certain subjects than about struggling to understand why the world of knowledge seemed already organized for me.

I was expected to learn formulas, theories, and concepts without always understanding why these particular representations had become the foundations of education. I wondered whether the world could have developed differently, but I did not yet understand how to turn that curiosity into a question about cognition.

Learning about representation and computation has given me a new perspective. Human knowledge can be understood as an ongoing interaction between the world, our cognitive capacities, and the representations we create. Existing representations make increasingly complex thought possible, while their structures also influence how later generations learn, reason, and discover.

This does not mean that truth is entirely determined by human representation. Nor does it mean that our current knowledge systems are merely accidental. Instead, it suggests that understanding the development of knowledge requires us to distinguish the constraints imposed by reality from those introduced by our cognitive tools and historical choices.

I still wonder whether another civilization, starting from a different intellectual history, would eventually arrive at something resembling our world. Perhaps it would discover many of the same truths while expressing them through representations we would barely recognize.

And perhaps that is what I find most fascinating about cognition: the possibility that reality is larger than any particular way we have developed to understand it.

I began by asking why I had to learn the knowledge that previous generations had created. Now, I find myself asking a different question.

If learning gives us the representations through which we understand the world, how can we also learn to think beyond them?

References

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